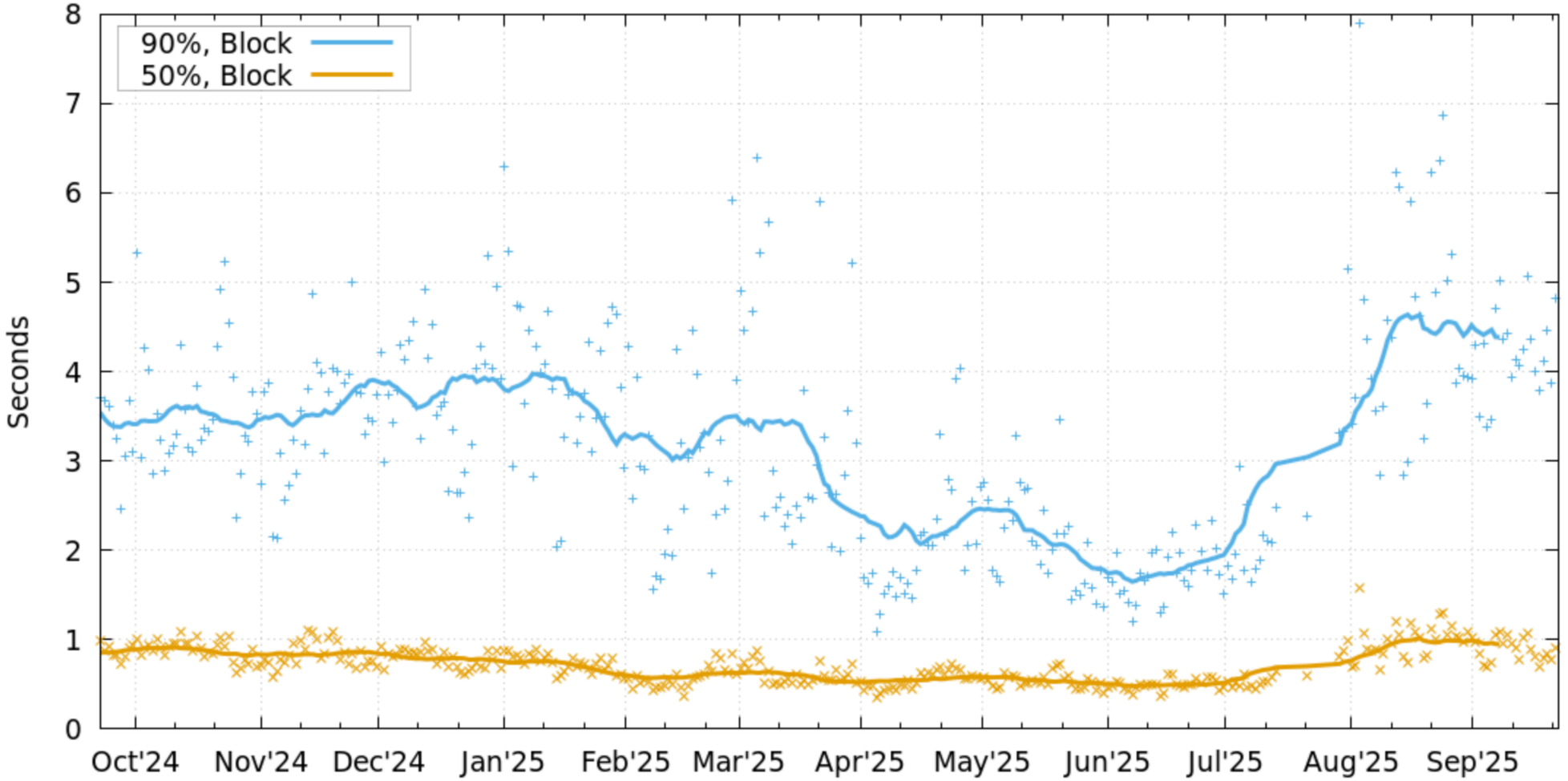
sub-1 sat/vbyte summer

@stratospher on 29/9/2025

Block Propagation Delay History



https://www.dsn.kastel.kit.edu/bitcoin/

mempool
@mempool

don't overpay for blockspace
0.1 sat/vB is the new 1 sat/vB



4:48 PM · Jul 15, 2025 · **171.9K** Views

Minimum relay feerate

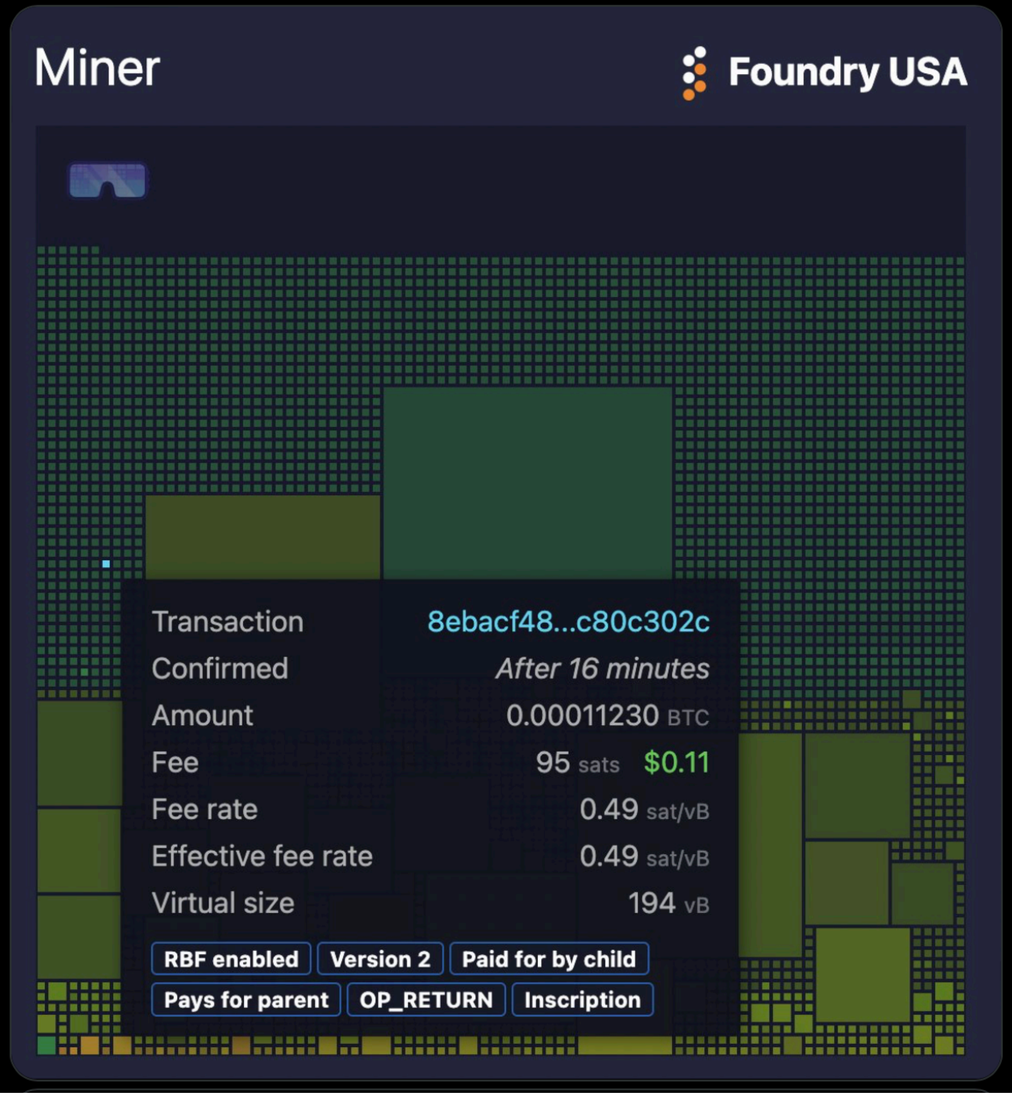
- The *minimum relay feerate* is the lowest fee per vByte (sat/vB) at which a node will accept and relay a transaction.

The minimum relay feerate is a DoS protection rule

representing a price on the network bandwidth used to relay transactions that have no PoW.

mononaut ✅
@mononautical

Guess who joined Sub-1 Sat/vB Summer?

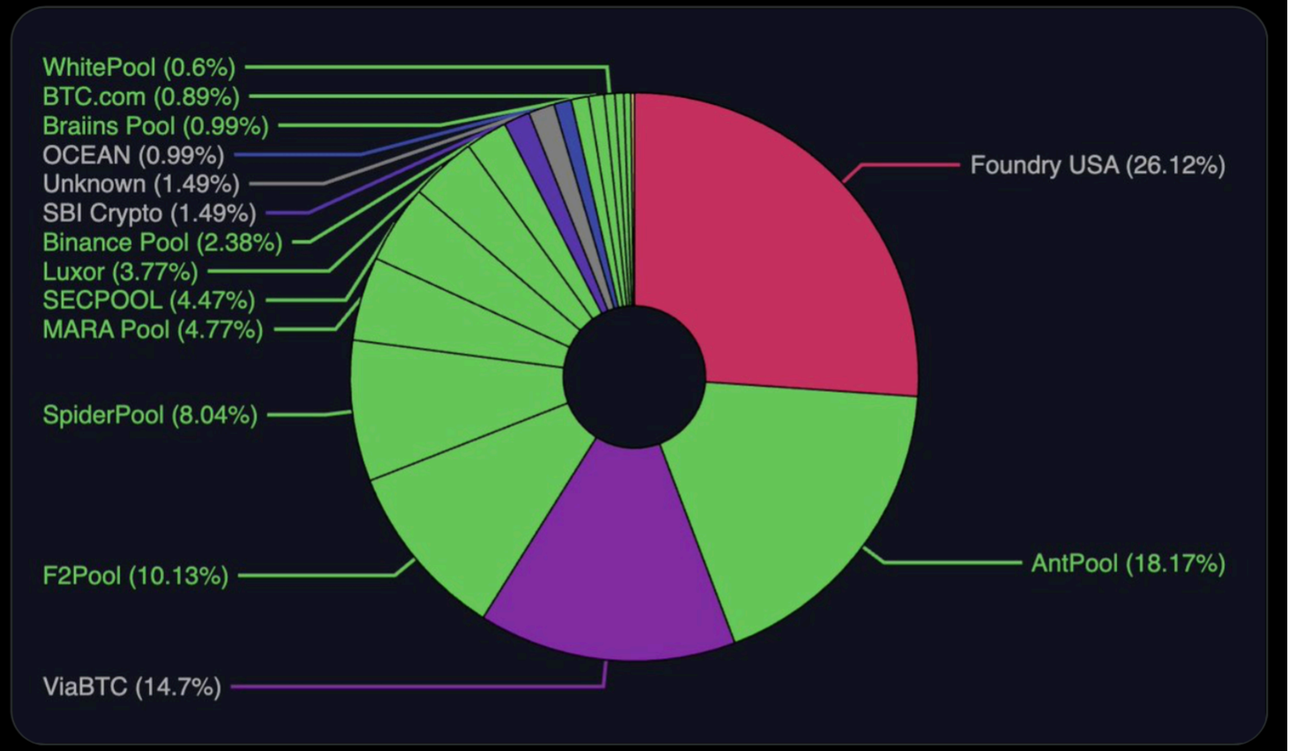


mononaut ✅
@mononautical

SECPOOL, Binance Pool, WhitePool, BTCcom and Mining Squared are all now mining sub-1 sat/vb blocks.

Braiins appears to be on board too judging by their stratum jobs, but just hasn't found any while fees were low enough.

Estimated hashrate for Sub-1 Sat/vB Summer is ~55%
x.com/mononautical/s...



Reduce minrelaytxfee to 100 sats/kvB #32959

Closed **RobinLinus** wants to merge 1 commit into `bitcoin:master` from `RobinLinus:reduce-minrelaytxfee`

Conversation 74 | Commits 1 | Checks 19 | Files changed 2



RobinLinus commented on Jul 13 • edited

Motivation

The default relay-fee (-minrelaytxfee) was set to 1000 sat/kvB about 10 years ago. Over that period the USD price of BTC has risen by roughly 2-3 orders of magnitude, making the economic cost of relaying small transactions disproportionately high.

Proposed change

Lower the default to 100 sat/kvB—a 10× reduction—bringing the effective USD cost back in line. This maintains DoS resistance without discouraging low-value UTXO consolidation.

47 1

Previous settings have been:

- 2015/10/13 - 5sat/vb to 1sat/vb
  - $250/BTC
  - 1sat*200vb = 0.5c
  - 1sat/vb*1Mvb/25BTC = 0.04% of block reward
- 2015/10/09 - 1sat/vb to 5at/vb
  - $250/BTC
  - 5sat*200vb = 2.5c
  - 5sat/vb*1Mvb/25BTC = 0.2% of block reward
- 2013/04/13 - 10sat/vb to 1sat/vb
  - $120/BTC [edit: found a different price source that doesn't seem as crazy]
  - 10sat*200vb = 0.024c
  - 10sat/vb*1Mvb/25BTC = 0.4% of block reward

Relay Cost Argument

- Relaying a transaction isn’t free — it consumes **bandwidth** across all nodes.
- Example:
  - 1 kB transaction × 100,000 nodes = **100 MB data transferred**.
  - Bandwidth cost (on AWS/EC2 in 2015) ≈ $0.10/GB → 1 cent for that relay.
  - At $250/BTC, 1000 sats = 0.25 cents. So 1 sat/vB (1000 sats for 1kB tx) roughly **covers relay cost**.
- Today:
  - Bandwidth costs haven’t changed much.
  - Node count hasn’t grown dramatically.
  - With BTC at ~$60k now, 1 sat/vB is **overpaying massively** relative to real relay costs.
  - Rough math: relay cost is closer to **0.01 sat/vB**.
  - So, **0.1 sat/vB would still cover relay even if node count increased 10x**.

Pros vs Cons

- Should Bitcoin nodes enforce a higher minimum fee to:
  - protect the network from spam and
  - set an expectation of meaningful fee pressure for long-term miner revenue?

OR

- Should they lower it to:
  - better align with actual network costs,
  - avoid propagation inefficiencies, and
  - allow legitimate low-value transactions?

How many peers relay sub-1 sat/vbyte?

Murch
@murchandamus

I was wrong. I did not account for inbound `block-only-relay` nodes not sending a `minfeefilter` message and therefore showing up as 0.
After removing nodes older than v0.13.0, and non-Bitcoin Core nodes, I think I have eight peers that accept sub-1-sat/vB.

```
"relaytxes": true, "subver": "/Satoshi:0.14.0/", "minfeefilter": 0.00000000,
"relaytxes": true, "subver": "/Satoshi:0.15.2/", "minfeefilter": 0.00000974,
"relaytxes": true, "subver": "/Satoshi:24.0.1/", "minfeefilter": 0.00000000,
"relaytxes": true, "subver": "/Satoshi:24.0.1/", "minfeefilter": 0.00000000,
"relaytxes": true, "subver": "/Satoshi:24.0.1/", "minfeefilter": 0.00000000,
"relaytxes": true, "subver": "/Satoshi:28.0.0/", "minfeefilter": 0.00000000,
"relaytxes": true, "subver": "/Satoshi:28.1.0/Knots:20250305/", "minfeefilter": 0.00000000,
"relaytxes": true, "subver": "/Satoshi:28.99.0/", "minfeefilter": 0.00000010,
```

> **Murch** @murchandamus · Sep 2
>
> If you think it's inconceivable that sub-1sat/vB transactions are organically relayed on the network, perhaps check what your node's peers accept:
>
> 48 of my 124 peers accept sub-1sat/vB txs.

```
Duna:~$ bitcoin-cli getpeerinfo | grep 'minfeefilter' | sort -nk2r | uniq -c | sor

5      "minfeefilter": 0.00001000,
6      "minfeefilter": 0.00000000,
1      "minfeefilter": 0.09170997,
1      "minfeefilter": 0.00000974,
1      "minfeefilter": 0.00000010,
```

11:08 AM · Sep 2, 2025 · **13.5K** Views

https://delvingbitcoin.org/t/measuring-minrelaytxfee-across-the-bitcoin-network/1989